RESEARCH PROJECT PROPOSAL

Toward a Cloud Forensic Readiness Framework for Bangladesh's Banking Sector: Bridging Evidence Acquisition and Cross-Border Jurisdictional Gaps

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Abstract

Cloud computing has become integral to Bangladesh's financial sector, yet the country's digital evidence law, forensic institutions, and international legal cooperation mechanisms have not kept pace with this shift. This proposal argues that Bangladesh's banking sector faces a specific and under-examined cloud forensic readiness gap: financial institutions operate under a 2023 Bangladesh Bank cloud computing guideline that regulates where data may be hosted but appears silent on how regulators or investigators can forensically access that data during an incident, while Bangladesh itself remains outside the Budapest Convention, the 2025 UN Convention against Cybercrime, and any confirmed CLOUD Act or US Mutual Legal Assistance Treaty arrangement. The result is that when cloud-hosted financial evidence is needed — as starkly illustrated by the 2016 Bangladesh Bank cyber heist — investigators must rely on ad hoc, low-success-rate cooperation with foreign cloud providers rather than an enforceable legal pathway. Building on a completed PRISMA-guided systematic review of the global cloud forensics literature, this project proposes to (1) analyze Bangladesh's regulatory, institutional, and jurisdictional constraints on cloud evidence acquisition in the banking sector specifically, (2) validate these constraints against practitioner input from forensic investigators and bank ICT security officers, and (3) design a cloud forensic readiness framework tailored to what is legally and technically achievable within Bangladesh's current constraints, rather than one that assumes access mechanisms Bangladesh does not have. The expected contribution is both academic — the first Bangladesh-specific cloud forensic readiness framework identified in the literature — and practical, offering Bangladesh Bank, CID's Digital Forensics & Cyber Police Centre, and individual financial institutions a concrete, implementable reference point.

1. Introduction and Background

Cloud computing offers scalability and cost efficiency, but it also disperses digital evidence across virtualized, provider-controlled infrastructure that investigators cannot physically access. A completed PRISMA-guided systematic review of 48 international sources (Kabir, 2026, unpublished survey) established that this problem manifests globally across four dimensions: evidence acquisition, evidence trustworthiness, legal and jurisdictional compliance, and standardization. That review, however, remained deliberately general — it surveyed the global state of the field without grounding the problem in any specific country, sector, or institution.

Bangladesh presents a case where these four global challenges converge with unusual force and specificity. Bangladesh has no in-country hyperscale cloud region — AWS, Microsoft Azure, and Google Cloud all serve Bangladeshi customers from offshore regions (Singapore, Mumbai, or elsewhere), with AWS's only in-country footprint being customer-premises Outposts hardware introduced in 2022. Bangladesh Bank, the central bank, responded to this reality with a 2023 cloud computing guideline (BRPD Circular No. 05) that regulates data classification and cross-border hosting approval for banks and financial service providers — but published summaries of this guideline do not address how a regulator or forensic investigator would obtain evidence from that cloud-hosted data during an investigation. Meanwhile, Bangladesh's own capacity to admit and process digital evidence is recent (the Evidence (Amendment) Act 2022) and its international cooperation position is thin: Bangladesh is not a party to the Budapest Convention on Cybercrime, did not sign the 2025 UN Convention against Cybercrime, and has no confirmed CLOUD Act agreement or bilateral US Mutual Legal Assistance Treaty, despite the fact that the major cloud providers serving Bangladeshi institutions are US-jurisdiction companies.

This gap is not hypothetical. The 2016 Bangladesh Bank cyber heist — in which attackers used the SWIFT payment network to attempt to steal roughly $1 billion, successfully transferring $81 million before the scheme was interrupted — remains Bangladesh's most consequential cybersecurity incident and has recently been re-analyzed in the international literature (Islam, 2026) as a case of institutional fragility in a developing-country financial system. It directly led to the creation of Bangladesh's national CERT, BGD e-GOV CIRT. Nearly a decade later, however, no Bangladesh-specific cloud forensic readiness framework has been proposed in the academic literature to address how such an incident's cloud-hosted evidence — logs, transaction records, communications — should be acquired, preserved, and made admissible when it may sit on servers outside Bangladeshi jurisdiction.

2. Problem Statement

Bangladesh's banking sector is adopting cloud computing under a regulatory framework (Bangladesh Bank BRPD Circular 05/2023) that governs data residency and hosting approval, but not forensic or regulatory access to that data during an incident. This regulatory silence exists alongside three compounding constraints: (a) Bangladesh has no in-country hyperscale cloud infrastructure, meaning most cloud-hosted financial data legally resides outside the country by default; (b) Bangladesh's domestic forensic capacity is documented as constrained, with a growing case backlog in CID's digital forensic labs (Barkat et al., 2020) and police officers reporting a need for substantially more training (Sharmin & Rahman, 2025); and (c) Bangladesh's international legal cooperation position leaves it dependent on ad hoc, non-treaty-based requests to foreign cloud providers, which UNODC itself describes as minimally utilized in practice, and which transparency reporting shows succeed only 64-68% of the time even when attempted (Meta, 2024). No existing framework in the literature — Bangladeshi or international — addresses cloud forensic readiness under this specific combination of regulatory, infrastructural, and jurisdictional constraints.

3. Research Gap

- No peer-reviewed literature was found proposing a cloud forensic readiness framework specific to Bangladesh, or to a developing-country banking sector operating without in-country hyperscale cloud infrastructure.
- Bangladesh Bank's 2023 cloud computing guideline has not been analyzed in the academic literature for its forensic-access implications specifically — existing commentary addresses it only from a data-residency and compliance perspective.
- Existing Bangladesh-specific digital forensics scholarship (Haque & Hossain, 2017; Sharmin & Rahman, 2025) addresses general digital forensics and predates or omits cloud-specific and cross-border evidence acquisition challenges.
- The 2016 Bangladesh Bank heist has been re-analyzed as an institutional-fragility case (Islam, 2026) but not specifically as a cloud/cross-border evidence-acquisition case, leaving its forensic-readiness lessons unexamined.
- The interaction between the new Personal Data Protection Ordinance 2025 (which appears to mandate local data mirroring) and the existing Bangladesh Bank cloud guideline has not yet been examined for its combined effect on forensic evidence acquisition.

4. Research Questions and Objectives

Research Questions

1. RQ1: What specific evidence-acquisition challenges does Bangladesh's banking sector face when conducting cloud forensic investigations, given the absence of in-country hyperscale cloud infrastructure and the requirements of Bangladesh Bank's 2023 cloud computing guideline?

2. RQ2: How do Bangladesh's jurisdictional position — non-membership in the Budapest Convention, absence from the 2025 UN Convention against Cybercrime, and lack of a confirmed CLOUD Act or US MLAT arrangement — affect the practical admissibility of cloud-hosted evidence under the Evidence (Amendment) Act 2022's Section 65B certificate requirement?

3. RQ3: What cloud forensic readiness framework can be designed that is achievable within Bangladesh's current regulatory, infrastructural, and institutional constraints, rather than assuming access mechanisms the country does not currently have?

Objectives

- To systematically map the regulatory, infrastructural, and jurisdictional constraints specific to cloud forensic evidence acquisition in Bangladesh's banking sector.
- To empirically validate these constraints through structured input from forensic practitioners (CID Digital Forensics & Cyber Police Centre) and financial-sector ICT security officers.
- To design and propose a cloud forensic readiness framework tailored to Bangladesh's banking sector, addressing evidence acquisition, chain-of-custody, and legal admissibility within existing constraints.
- To evaluate the proposed framework against the 2016 Bangladesh Bank heist as a retrospective validation case.

5. Literature Review (Summary)

This proposal builds on two literature bases, detailed fully in the accompanying Literature Summary Report. The first is the international cloud forensics literature already synthesized in a completed PRISMA-guided systematic review (48 sources), which established a four-dimensional taxonomy of evidence acquisition, evidence trustworthiness, legal/jurisdictional compliance, and standardization, and identified explainable AI, blockchain-based evidence integrity, and forensic-readiness-by-design as active research directions. That review provides the technical and conceptual vocabulary this proposal applies to Bangladesh specifically.

The second, newly assembled literature base is Bangladesh-specific, spanning: primary legislation and regulation (the Evidence (Amendment) Act 2022, the Cyber Security Act 2023, the Personal Data Protection Ordinance 2025, and Bangladesh Bank's cloud computing and ICT security guidelines); institutional and capacity assessments (Barkat et al., 2020; Sharmin & Rahman, 2025; Uzzal, 2025); analyses of the 2016 Bangladesh Bank heist as a financial-sector cybersecurity case (Islam, 2026; Mazumder & Sobhan, 2020; Lee & Mautz, 2025); and documentation of Bangladesh's jurisdictional position on international cybercrime cooperation (Council of Europe Octopus project; UNODC, 2024; Meta transparency reporting, 2024). Notably, no source in either literature base proposes a cloud forensic readiness framework specific to Bangladesh or to a comparable developing-country banking context, which is the gap this proposal addresses.

6. Proposed Methodology

Phase 1 — Regulatory and Documentary Analysis (Weeks 1-3)

Direct analysis of primary legal and regulatory texts — Bangladesh Bank BRPD Circular 05/2023, the ICT Security Guideline v4.0, the Evidence (Amendment) Act 2022, the Cyber Security Act 2023, and the Personal Data Protection Ordinance 2025 — to confirm and document the specific provisions (or silences) relevant to cloud forensic evidence acquisition. This directly tests the hypothesis, raised in the literature review, that the Bangladesh Bank cloud guideline does not address forensic or regulatory audit access.

Phase 2 — Practitioner Validation (Weeks 4-6)

Semi-structured interviews or a structured questionnaire administered to a small purposive sample of forensic practitioners (CID Digital Forensics & Cyber Police Centre, BGD e-GOV CIRT) and bank ICT security or compliance officers, to validate the constraints identified in Phase 1 against practical experience and to surface constraints not visible in documentary analysis alone. This follows the methodological precedent of Sharmin & Rahman (2025), who surveyed Bangladesh Police officers on digital forensics capacity.

Phase 3 — Framework Design (Weeks 7-9)

Using a design-science research approach, synthesize the findings from Phases 1-2 with the four-dimensional taxonomy from the completed international literature review to design a cloud forensic readiness framework specific to Bangladesh's banking sector. The framework will explicitly address what evidence acquisition mechanisms are legally and practically available under current constraints, rather than assuming mechanisms (such as CLOUD Act cooperation) that Bangladesh does not currently have access to.

Phase 4 — Retrospective Validation (Weeks 10-11)

Apply the proposed framework retrospectively to the 2016 Bangladesh Bank heist as a case study, using the incident timeline and institutional response documented in existing analyses (Islam, 2026; Lee & Mautz, 2025) to assess whether the framework would have improved evidence acquisition, chain-of-custody, or cross-border cooperation outcomes had it existed at the time.

Phase 5 — Write-Up (Week 12)

Compilation of the complete research article, incorporating the regulatory analysis, practitioner validation findings, proposed framework, and retrospective case study evaluation.

7. Expected Contribution and Significance

- Academic: the first cloud forensic readiness framework identified in the literature that is specific to Bangladesh, and one of few addressing a developing-country banking sector operating without in-country hyperscale cloud infrastructure.
- Regulatory: a concrete, evidence-based assessment of whether Bangladesh Bank's cloud computing guideline addresses forensic access, directly usable by Bangladesh Bank in any future revision of the guideline.
- Institutional: a practical reference framework for CID's Digital Forensics & Cyber Police Centre and BGD e-GOV CIRT when handling cloud-related financial-sector incidents.

- National relevance: directly responsive to Bangladesh's stated policy priorities under Digital Bangladesh/Smart Bangladesh initiatives around data-driven governance and cybersecurity capacity, and to the documented, ongoing gap in Bangladesh's international legal cooperation on cybercrime.

8. Scope and Limitations

This proposal is scoped to Bangladesh's banking and financial services sector specifically, rather than cloud forensics in Bangladesh generally, in order to maintain a tractable and well-evidenced focus. Government e-services and other sectors that also use cloud infrastructure are acknowledged as related but out of scope. Practitioner validation (Phase 2) will rely on a purposive, not statistically representative, sample, consistent with the exploratory nature of this early-stage framework design; the findings should be understood as framework-informing rather than population-generalizable. Primary regulatory texts (the Bangladesh Bank cloud guideline in full, and the Personal Data Protection Ordinance 2025 gazette text) will be obtained and read directly during Phase 1 rather than relied upon secondhand, and any discrepancy between secondary reporting and the primary text will be corrected and documented transparently.

9. Work Plan and Timeline

Phase	Activity	Weeks
Completed	PRISMA-guided systematic review of global cloud forensics literature (48 sources); four-dimensional taxonomy; comparative analysis	Prior to this proposal
1	Regulatory and documentary analysis of Bangladeshi legal/regulatory texts	1-3
2	Practitioner validation (CID, BGD e-GOV CIRT, bank ICT security officers)	4-6
3	Framework design (design-science synthesis)	7-9
4	Retrospective validation against the 2016 Bangladesh Bank heist	10-11
5	Full research article write-up	12

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Note: A complete list of the 48 references from the antecedent global cloud forensics systematic review is provided in the accompanying survey paper and is not repeated here.